PALEOMAGNETIC INVESTIGATIONS ON THE EAST AFRICAN RIFT IN NORTHERN TANZANIA

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ABSTRACT

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Near Lake Manyara and north of Monduli, Northern Tanzania, Precambrian gneisses and Neogene basalts on the East African rift were sampled. Rock magnetic and microscope investigations established coarse and fine grained magnetite as carrier of the NRM in the basalts, and hematite in the gneisses. In all three sections only reversed magnetization directions were found. The Precambrian gneisses show very hard magnetizations which is known to be prefolding and possibly of Proterozoic age.

INTRODUCTION

Within a mineral prospecting program in Northern Tanzania, sponsored by UNDP and Tanzania, the possibility arose to take paleomagnetic samples on the eastern branch of the East African Rift. The two major targets were to test the suitability of different rock types like Precambrian gneisses and Neogene basalts for paleomagnetic purpose and the application of reversal stratigraphy as an aid to interpreting the geology. Three sections were chosen. The first one near Lake Manyara runs down the small Marera Valley north of Manyara hotel. From the top to the bottom, where the talus deposits start, one can distinguish 16 layers. Sometimes they are separated by thin (0,5 m) tuffite layers. The thickness of the basalt flows varies between 3 and 10 meters.

Stratigraphically, they belong to the informally named "Older Extrusives" from the lowermost part of the Noegene (R. Pickering, 1963). Absolute age data are not available so far. The rocks are basaltic lavas, sometimes olivine basalts. Under the microscope, the olivines show strong weathering effects as evidenced by brown edge zones. West of this section, carbonatites and

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Precambrian gneisses were sampled near Oldoinyo Dili. The acid gneisses are mainly built up of the minerals quartz, microline, albite or oligoclase and biotite; sometimes with orthoclase.

Tectonically the area is characterized by volcanoes and block faulting, the main feature being the Gregory rift. The tectonism is thought to be of Plio-Pleistocene age, with a slight W-tilting taking place post mid-Pleistocene.

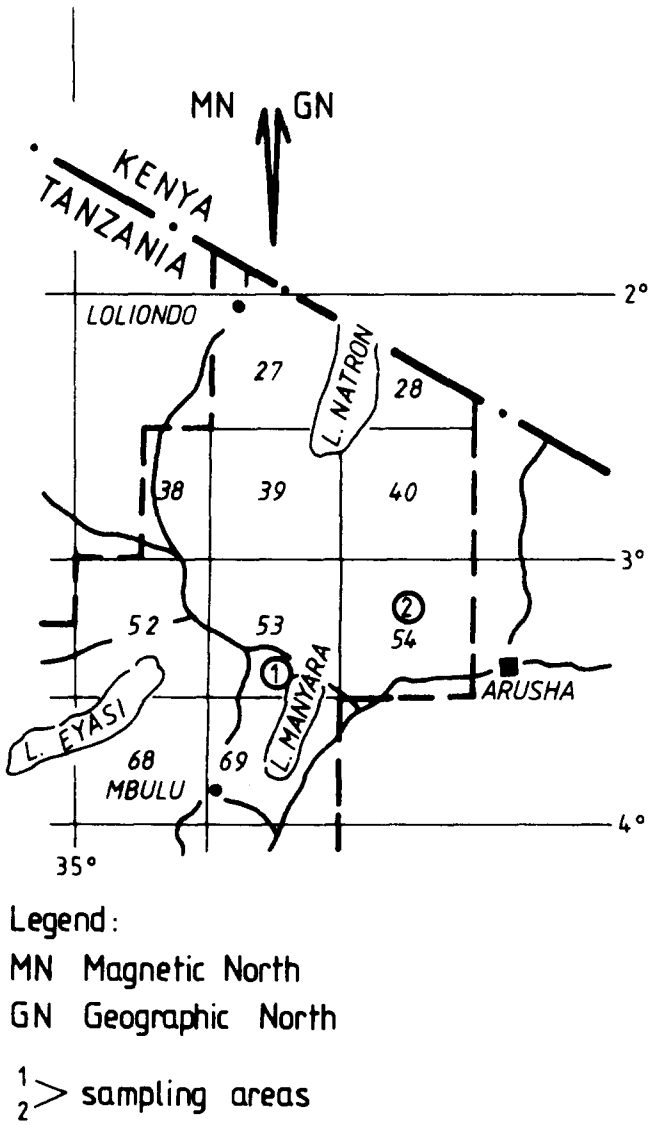


Fig. 1. Geographic location of the investigated areas.

The two sections N of Monduli are nearly opposite each other. The W-section was taken on the road from Monduli to Lake Natron where it crosses the Tarosero Scarp, whereas the eastern section runs down the Matunginni Scarp. Both sections are in the olivine basalts, which again belong to the "Older Extrusives" (Dawson, 1961). The individual basalt flows in this area are always separated from each other by basaltic tuffs. Rifting in this area is also thought to be of Plio-Pleistocene age.

ROCK MAGNETIC EXPERIMENTS

From the samples, which were taken by hand drills, a large number of pilot samples were selected, representing the great variety of rock types. The pilot samples were selected to investigate their suitability for paleomagnetic purposes. Thermal and alternating magnetic field (AF) cleaning, isothermal remanent magnetization (IRM) aquisition and saturation magnetization versus temperature (J_s-T) experiments were carried out, establishing magnetite as the main carrier of the natural remanent magnetization (NRM). The observed blocking temperature was nearly always around 570°C (Fig. 2a-d). Figure 2a shows the data for the carbonatites. In the vector diagram, three components of remanence directions can be seen, a stable end direction being obtained above 400°C. As the outcrops are dislocated, they are not discussed any further. In the Manyara section a very soft component or viscous remanent magnetization (VRM) was noticed, and above 50°C nearly 90% of the NRM intensity was lost. This soft component is thought to be due to ironhydroxide. The Zijderveld diagrams in Fig. 2a-d show two components, which demonstrates successful thermal cleaning. AF-cleaning also shows a removal of the soft secondary component up to 100 Oe. Above this field, a straight line decay towards the origin is obtained (Fig. 4b).

The IRM-aquisition curves show saturation between 0,2 and 0,3 T for the representative samples (Fig. 3), which clearly show the presence of magnetite. The J_S-T curves support the microscope analyses showing a broad scatter of magnetite grain sizes. The susceptibility curves demonstrate chemical stability during heating. The gneisses, which are weathered, show a strong increase in the intensity between 50 and 100°C. Afterwards their behaviour is very stable up to 500°C, after which the intensity starts to decrease to a final blocking temperature of about 650°C. This behaviour is interpreted as due to two phase mineralization, where the soft phase is due to Ironhydroxide, and the hard one to hematite. The Zijderveld plots are consistant with a 2-phase mineralization since the soft component is more or less antiparallel to the hematite direction. This results from the

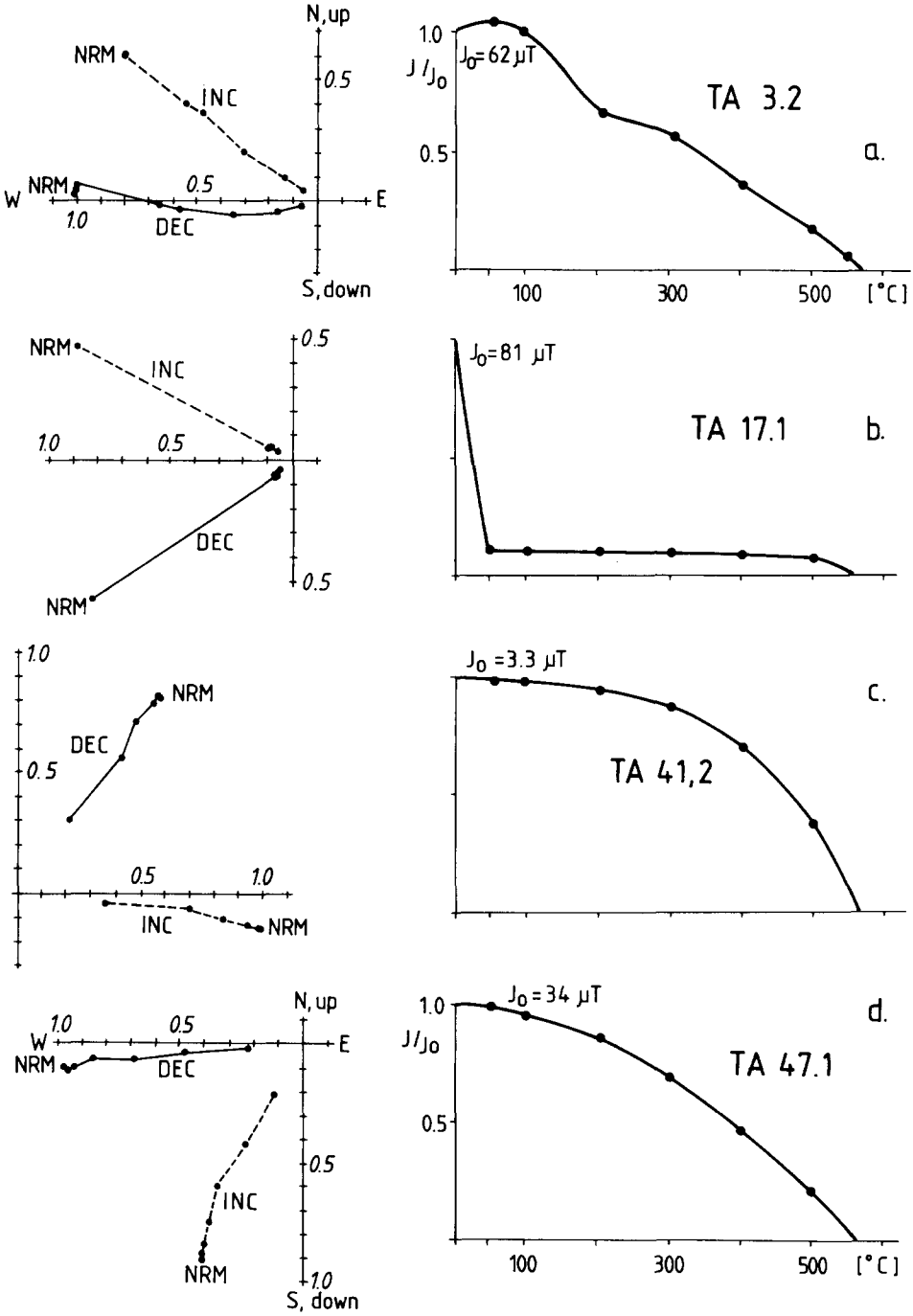


Fig. 2. Rock magnetic behaviour of pilot samples.

a. Carbonatite of Oldoinyo Dili

b. Basalt of Lake Manyara

c. Olivine basalt of Matunginni Scarp

d. Olivine basalt of Tarosero Scarp

These pilot results are shown in Zijderveld diagrams and thermal-demagnetization curves. The Zijderveld plots show DEC and INC versus normalized intensity, with INC in the vector plane.

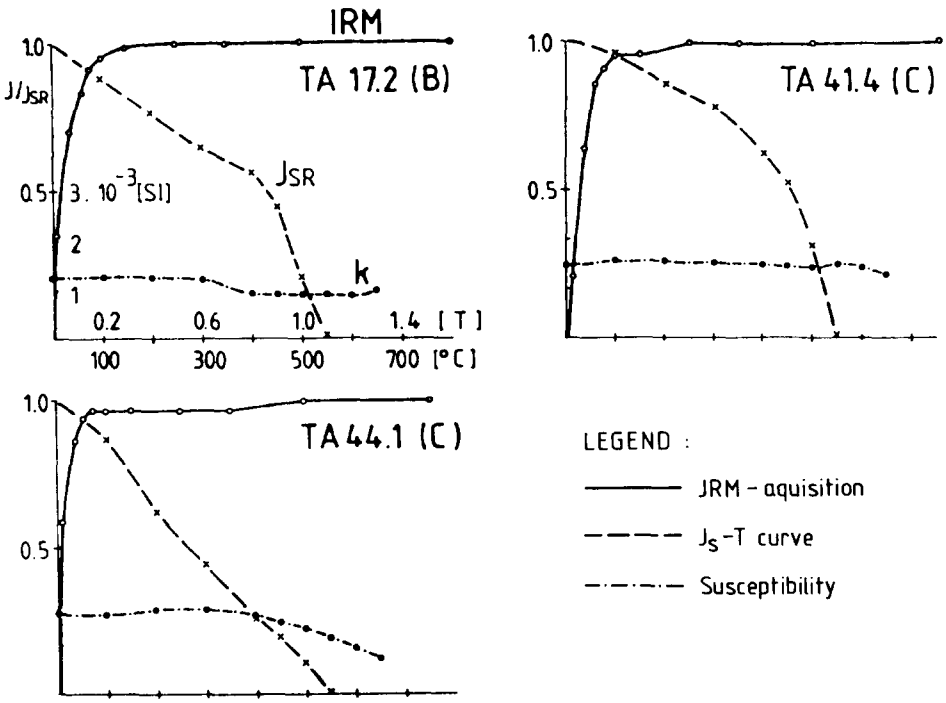


Fig. 3. IRM-aquisition, J_{SR}-T and susceptibility (k) curves. The normalized IRM is shown versus progressive DC-field, the normalized saturation magnetization (J_{SR}) versus stepwise increasing temperature, the susceptibility (in SI-units) after each step of thermal treatment and cooling down in a field free space.

dehydration and transformation of the ironhydroxide decreasing its influence. After 100 to 150°C only hematite is left with a demagnetization line towards the origin.

PALEOMAGNETIC RESULTS AND INTERPRETATION

For all samples from the rift area, the cleaning was done very carefully and in different steps, to work out,where ever possible, changing magnetization polarities and best fitting characteristic remanent magnetization (ChRM) directions. As shown in Fig. 5 the gneisses, were cleaned successfully.

In the previous section it was shown that the gneisses have at least two, and possibly three components of remanent magnetization. A very soft one due to ironhydroxide with a direction close to the present field direction (Hicken *et al.*, 1972). The second component could be caused by a weak

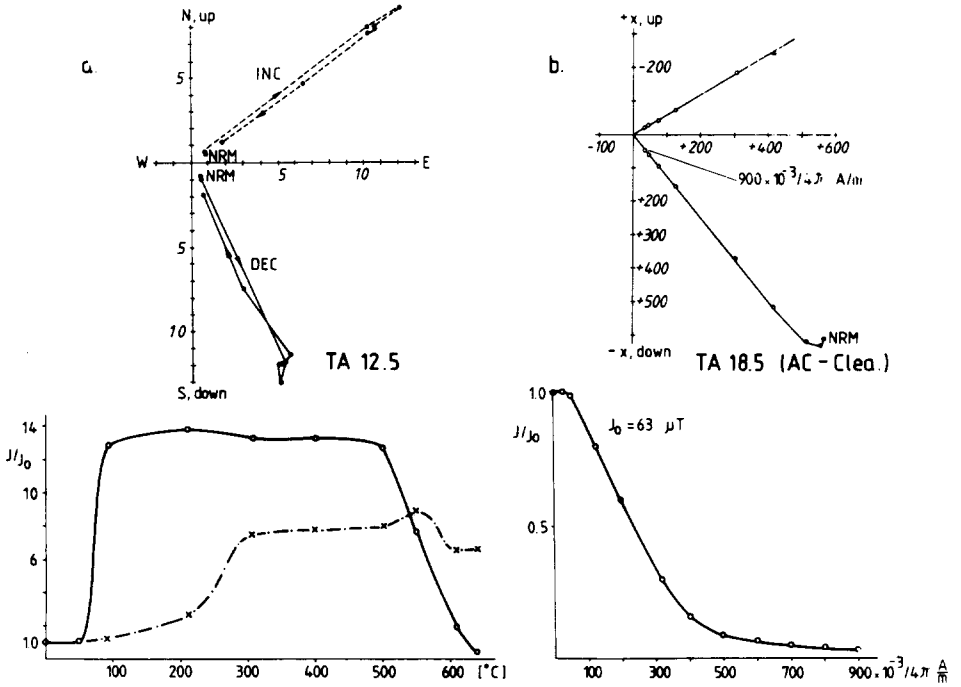


Fig. 4. Cleaning results

a. Thermal cleaning behaviour of precambrian gneisses. The Zijderveld-plot shows Declination and Inclination versus normalized intensity (NRM) with Inclination in the vector plane. In the thermal treatment diagram the full line shows the normalized intensity (NRM) and the dotted line the susceptibility after each step of demagnetization in SI-units.

b. AC-cleaning of a basalt sample from Lake Manyara. The Zijderveld-plot shows the behaviour of a basalt pilot sample (TA-18.5) during stepwise increased AC-field, declination is projected in the y-z plane. Below, the normalized intensity versus AC-field strenght is shown.

partial thermal overprint (PTRM) of the hematite during the pre-Neogene carbonatite intrusion in the Oldoinyo Dili area. Finally, above 550°C, there is a stable end direction. This direction is most likely the direction aquired during the metamorphism. Even though there is no possibility of a proper foldtest one can see from Fig. 5, that this end direction could represent a prefolding magnetization direction. Furthermore, if one agrees with the results of Krs (1983) from a more southern part of N-Tanzania, the metamorphism took place while the sediments were in a more or less original position, then the bedding parameters may be used for tectonic corrections. Comparing this poleposition with the apparent polar wander path (APWP) for the African cratons (McElhinny & McWilliams, 1977, Piper, 1982) one gets a reosonable fit at about 2300 Ma. This lower proterozoic result agrees with other results for the Congo craton within

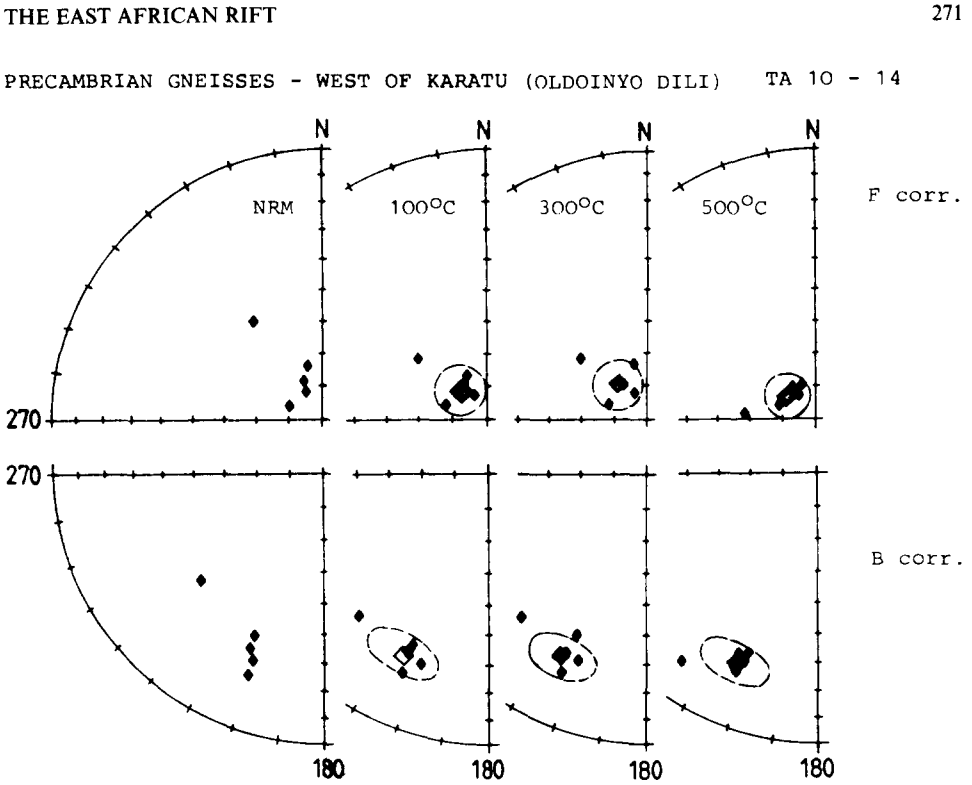


Fig. 5. Paleomagnetic results of the precambrian gneisses. The squares are site means (NRM, 100°, 300°, and 500°C) before (F-corr) and after tectonic (B-corr) correction.
The site mean directions of the magnetically cleaned remanence vectors are shown on an equal area projection.The overall mean direction is shown with big squares and cone of confidence (α_{95}).

$\pm$100 Ma (Piper, 1982; Krs, 1983). This interpretation will be tested during the next sampling in 1984.

A magnetic overprint aquired during the time of Pleistocene volcanism seems to be impossible, as the directions before tectonic correction (F-corr) shows no correlation with Neogene directions. The Neogene basalts, as mentioned before, show a magnetically soft behaviour, which is the probable cause of the partly unstable results of the sampled sections.

Starting off with the Manyara section, one can see that the best grouping of site means is reached at 100°C, after eliminating the influence of ironhydroxide. At 300°C the scatter increases and at 500°C just 8 sites remain more or less stable (Fig. 6). This stability is seen in the Zijderveld plots (Fig. 2).

The Matunginni Scarp section is the only one which shows a successful cleaning with a good fitting average direction above 500°C. Just 3 sites increase in scatter and reduce the quality of the mean direction shown in Fig. 6.

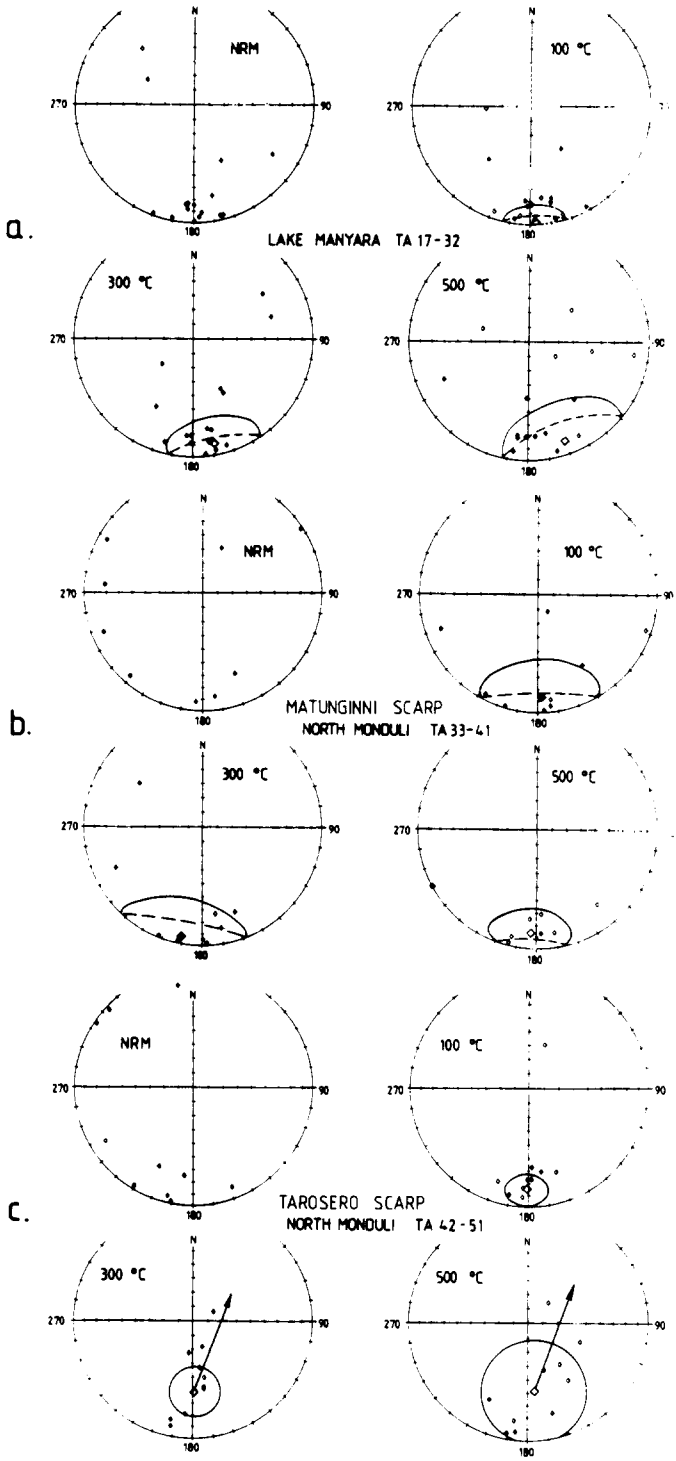

a.
NRM
100 °C
LAKE MANYARA TA 17-32
300 °C
500 °C
b.
NRM
100 °C
MATUNGINNI SCARP
NORTH MONDULI TA 33-41
300 °C
500 °C
c.
NRM
100 °C
TAROSERO SCARP
NORTH MONDULI TA 42-51
300 °C
500 °C

TABLE I

Final results from N. Tanzania.

Rock type	SITE Lat.	Long.	PALEO REMANENCE DIRECTION Treatm.	P_{Dec}	P_{Inc}	N	k	α_{95}	VGP P_{lat}	P_{long}
Precambrian Gneiss	−3.43	35.72	500°C	206.5	27.7	5	82.8	8.5	−61.55	330.76
Basalt Lake Manyara	−3.42	35.72	500°C	172.0	+14.4	8	13.3	15.8	−81.13	279.83
Basalt Matunginni S.	−3.10	36.48	500°C	183.2	−15.4	5	27.6	14.8	−78.60	232.72
Basalt Tarosero S.	−3.20	36.47	500°C	186.3	−4.0	4	8.9	32.6	−81.83	266.99

In contrast, the Tarosero Scarp section appears successfully cleaned at 100°C and increases in scatter above that temperature. This behaviour seems to be similar to that seen in the Manyara section.

As the pilot samples (Fig. 2a-d) show, there is no indication of a complicated multi-component magnetization. After eliminating weathering effects, a good grouping is obtained, which is thought to be the result of a PTRM from one of the last thermal events during the Pleistocene tectonism. The scatter of vector direction above this temperature is explained by relative motion between different parts of the sampled profile. This idea is supported by the direction in which the points scatter. They scatter towards a direction more or less perpendicular to the rift (Fig. 6). Whether this is due to recent downward creeping or bending during rifting or both, is difficult to decide.

CONCLUSION

As far as the Precambrian is concerned, the end direction fits the Proterozoic part of the polar wander path of McElhinny & McWilliams

Fig. 6. From top to bottom the 4 stages of NRM, 100°C, 300°C and 500°C for all site means of the three rift sections are shown. For the paleo-pole calculation, only the stable remaining sites were used. The arrow in 6.c gives the direction perpendicular to the rift axis.
The mean vectors of the sites are plotted on an equal area projection with the cone of confidence (α_{95}) after the above noted thermal treatment. Open symbols show inclinations above, full below horizontal. The squares show the mean vector directions of the rift sections.

(1977), and Piper (1982) quite well, and supports the idea of a unique tectonic history of the African cratons during the Precambrian. The basalt sections show a strong influence of relative motion between parts of the sections during rifting, except for the Matunginni Scarp section. There the scarp is more steep, probably less dislocated. None of the basaltic profiles show a normal magnetization direction. The paleo-polepositions (Table I) are in good agreement with previous poles in the lists of Hicken *et al.* (1972) Irving and Hastie (1975).

For the calculation of the paleo-poleposition, only the stable sites were used.

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REFERENCES

Dawson, J. B., 1961. Geological Survey of Tanzania, Quarter Degree Sheet 53: 1964.

Hicken, A., Irving, E., Law, K. L., Hastie, J., 1972. Catalogue of paleomagnetic directions and poles, First Issue, Vol. 45-No.1, Department of energy, mines and resources, Ottawa.

Irving, E. and Hastie, J., 1975: Caralogue of paleomagnetic directions and poles, Second Issue. Geomagnetic Series No.3, Ottawa.

Krs's, M., 1983. Paleomagnetic, Petromagnetic and Paleotectonic Investigations of Proterozoic Gabbro Body at Garauja-Basotu, Northern Tanzania.

McElhinny, M. W. and McWilliams, M. O., 1977. Precambrian geodynamics - a paleomagnetic view, Tectonophysics, 40,: 137–159, Amsterdam.

Pickering, R., 1963. Geological Survey of Tanzania, Quarter Degree Sheet 54: 1965.

Piper, J. D. A., 1982. The Precambrian paleomagnetic record; the case for the Proterozoic Supercontinent. Earth and Planetary Science Letters 59: 61.